Intelligent Patient Risk Assessment System Using Machine Learning and Advanced AI

Abstract

Heart disease remains one of the leading causes of mortality worldwide. Early risk prediction can significantly improve preventive healthcare and patient outcomes. This project presents an intelligent healthcare risk assessment system that predicts patient heart disease risk using classical machine learning techniques combined with advanced AI technologies.

The system performs data preprocessing, feature engineering, and model training using Logistic Regression, Decision Tree, and Random Forest classifiers. In addition, modern AI techniques such as LangChain, Retrieval-Augmented Generation (RAG), and Agentic AI are integrated to provide intelligent insights, report analysis, and automated decision-making.

A Streamlit-based interactive dashboard enables real-time patient data input, visualization, report generation, and AI-driven explanations. The system selects the best-performing model using evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. This solution provides a fast, reliable, and user-friendly tool for supporting early diagnosis and preventive healthcare decisions.

1 Introduction

Cardiovascular diseases are a major global health concern and are responsible for a significant number of deaths each year. Early detection and risk assessment can play a crucial role in reducing mortality rates.

Traditional diagnostic approaches require expert analysis and can be time-consuming. With the advancement of machine learning and artificial intelligence, automated systems can assist in predicting disease risk efficiently.

This project aims to build an intelligent system that combines machine learning with modern AI techniques to provide accurate predictions, explanations, and enhanced user interaction.

2 Problem Statement

Manual evaluation of patient health risk is often slow, subjective, and prone to human error. There is a need for an automated system that can analyze clinical data quickly and provide reliable predictions along with meaningful insights.

3 Objectives

- Develop a machine learning model to predict heart disease risk.
- Perform efficient data preprocessing and feature engineering.
- Compare multiple ML models to select the best performer.
- Integrate advanced AI techniques such as LangChain and RAG.
- Enable report upload, analysis, and generation.
- Provide a user-friendly and interactive interface.

4 Dataset and Preprocessing

The system uses a heart disease dataset containing clinical parameters such as age, cholesterol level, blood pressure, and heart rate.

4.1 Preprocessing Steps

- Missing values were handled using median imputation.
- Categorical values were encoded into numeric format.
- Data normalization was applied where required.
- Outliers were analyzed and treated.
- Target variable was converted into binary classification:

$$Target = \begin{cases} 1 & \text{if disease risk present} \\ 0 & \text{otherwise} \end{cases}$$

5 Methodology

5.1 Data Splitting

$$\text{80\% Training Data} \quad \text{20\% Testing Data}$$

5.2 Machine Learning Models

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

5.3 Feature Scaling

StandardScaler was applied to normalize features for Logistic Regression.

5.4 Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

6 Model Evaluation

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

7 Advanced AI Integration

7.1 LangChain

LangChain is used to integrate Large Language Models with the machine learning pipeline.

- Converts predictions into human-readable explanations
- Uses prompt templates for structured responses
- Maintains conversational context
- Connects multiple components into a pipeline

7.2 Retrieval-Augmented Generation (RAG)

RAG enhances system performance by combining retrieval and generation.

- Retrieves relevant medical knowledge
- Provides context-aware AI responses
- Reduces hallucination
- Improves reliability of explanations

7.3 Agentic AI

Agentic AI enables intelligent decision-making.

- Understands user intent
- Chooses appropriate task (prediction, analysis, report)
- Automates workflow execution

7.4 AI-Based Report Analysis

- Extracts data from uploaded medical reports
- Summarizes important findings
- Identifies risk indicators
- Combines report data with ML prediction

7.5 End-to-End Workflow

$$\text{User Input} \rightarrow \text{Data Processing} \rightarrow \text{ML Model} \rightarrow \text{LangChain} \rightarrow \text{RAG} \rightarrow \text{Agent} \rightarrow \text{Explanation} \rightarrow \text{Report}$$

8 System Architecture

$$\text{User Input} \rightarrow \text{UI Dashboard} \rightarrow \text{ML Model} \rightarrow \text{AI Layer} \rightarrow \text{Prediction} \rightarrow \text{Visualization} \rightarrow \text{Report}$$

9 User Interface

- Input patient data
- Select model
- View predictions
- Upload reports
- Download analysis
- View history

10 Additional Features

10.1 History Tracking

Stores previous predictions and user interactions.

10.2 Report Upload

Allows medical report analysis using AI.

10.3 Report Generation

Generates downloadable reports with predictions and explanations.

11 Results

Risk Score Range	Risk Level
0.00 – 0.39	Low Risk
0.40 – 0.69	Moderate Risk
0.70 – 1.00	High Risk

Table 1: Risk Classification

12 Visual Outputs

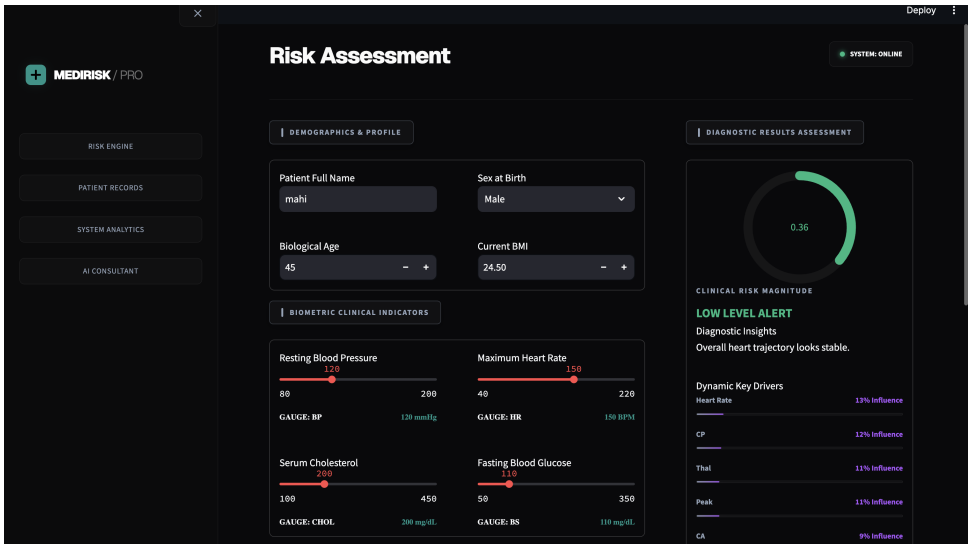

Figure 1: Streamlit Dashboard Interface

13 Advantages

- Fast predictions
- AI-powered insights
- Easy to use
- Real-time analysis

14 Limitations

- Depends on dataset quality
- Not a replacement for doctors
- Limited features

15 Conclusion

The project successfully integrates machine learning with advanced AI techniques to create an intelligent healthcare system. It provides accurate predictions, meaningful insights, and improved usability, making it a valuable tool for early risk detection and preventive healthcare.

16 References

- World Health Organization
- Scikit-learn Documentation
- Streamlit Documentation
- UCI Machine Learning Repository